When Does Graph Structure Help Noisy-Label Intrusion Detection? A Leakage-Audited Study of Rare-Class Training-Signal Retention

Abstract

Machine-learning intrusion detection is trained on imperfect labels, and standard noisy-label pipelines delete suspicious samples—often the rare-class rows that carry most of the learnable signal. We report a leakage-audited boundary study of when graph structure genuinely helps noisy-label network intrusion detection. Starting from a graph extension of collaborative label denoising (CoLD), we froze an audited protocol— label-construction proxy fields masked, strongest-available group-disjoint splits, pre-registered hypothesis families with Holm correction—and re-evaluated everything inside it on CICIDS-2017, UNSW-NB15, and the CESNET-TLS-Year22 service-classification stress test. Three findings survive. First (robust; 9/9 corrected tests): within a shared pipeline, soft retention improves tail-class F1 over hard deletion (+0.21 to +0.31 on CICIDS-2017; +0.03 to +0.05 elsewhere) at no detectable aggregate cost, and the direction survives real labelling errors and a 100k-window scale check. Second (null; reported with effect ceilings): a multi-view label-space graph diagnostic adds no detectable increment over Confident Learning (18/18 tests; Tail-F1 ceiling +0.026), and the specific rescue weighting adds nothing over a simple frequency floor. Third, a four-factor leakage decomposition attributes 25–36 points of Tail-F1 inflation to duplicates, cross-split entities, proxy fields, and clean-label oracle paths—factors

that flatter deletion-style baselines and compress the apparent advantage of retention. The audit also yields dataset-level findings: CESNET's released features contain direct label parents; UNSW-NB15's official partition ships with 38.4% exact train–test feature collisions; and ∼95% of CICIDS-2017's rare "Web Attack – Brute Force" class are labelling artefacts. The protocol, regression tests, and reproduction package accompany the paper.

Keywords: intrusion detection, noisy labels, graph learning, training-signal retention, leakage audit, rare-class robustness, negative results

1 Introduction

Machine-learning classifiers underpin modern network intrusion-detection systems (IDS) and security operations centers (SOCs). These models are trained from labels that are seldom perfect: they may be delayed, produced by rules of uneven quality, or shaped by analyst workflows that emphasize high-severity cases [2, 3]. The resulting label noise degrades both detection quality and operational triage. Crucially, the cost of label noise is asymmetric. Discarding a mislabeled benign flow may raise a benchmark score, but discarding an informative low-frequency sample can remove much of the training signal available for a rare class.

This asymmetry motivates a reframing of the problem. Conventional noisy-label pipelines treat suspicious samples as contamination to be purged; we instead treat data curation as a *training-signal retention* problem, in which the decision to delete weighs estimated label unreliability against the class-level cost of losing scarce training examples. This is a mechanism-level learning claim: benchmark rows are not, by themselves, SOC investigation or forensic evidence. The question is therefore not only *how accurately* noise can be detected, but *what should be done* with samples the detector cannot confidently classify: delete, trust, or retain at reduced influence.

CoLD [1] recently framed noisy-label IDS through *local consistency*: because different traffic classes can share feature neighborhoods, label noise induces spurious associations, and a collaborative representation can identify and remove likely-noisy samples before training. CoLD, like most dataset-purification methods, ultimately performs *hard deletion*. Starting from a graph-based extension of this idea, we ask three questions that turn out to have very different answers. (RQ-A) Does replacing hard deletion with signal-preserving soft weighting retain rare-class training signal and improve tail-class detection? (RQ-B) Does a multi-view graph diagnostic, kept strictly in label space, add measurable value over precise probabilistic pruning? (RQ-C) How much of the apparent benefit of graph-based denoising on standard IDS benchmarks is real, and how much reconstructs the rules by which the benchmark labels were generated in the first place?

Answering these questions honestly required an unusual amount of methodological hygiene, and the audit trail became a contribution in its own right. Our first implementations produced spectacular gains that did not survive scrutiny: a leakage audit revealed 24% exact-duplicate CICIDS-2017 records, clean-label paths in score and graph construction, label-construction proxy fields in the released features, and train–test entity overlap. We therefore froze an audited protocol—proxy-masked features, strongest-available group-disjoint splits, pre-registered hypothesis families with Holm correction—and re-ran everything inside it. Under that protocol the answers are: soft retention robustly preserves tail signal relative to deletion (RQ-A, positive); the graph diagnostic adds no detectable increment over Confident Learning, and the specific weighting form adds nothing over a frequency floor (RQ-B, null, with informative effect ceilings); and a four-factor decomposition attributes 25–36 points of Tail-F1 inflation to leakage factors that flatter deletion-style methods most (RQ-C).

Contributions.

The empirical record supports the following bounded contributions:

1. A *four-factor leakage audit protocol* for noisy-label IDS (duplicates, cross-split entities, label-construction proxy fields, clean-label oracle paths), instantiated as frozen splits, masking tiers, and regression tests. It quantifies 25–36 points of Tail-F1 inflation, shows that leakage flatters deletion-style baselines and compresses the apparent advantage of retention, and surfaces three dataset-level findings: CESNET's released `CATEGORY` columns are direct label parents, UNSW's official partition ships with 38.4% exact feature collisions, and CICIDS's flow topology makes entity-disjoint splitting infeasible by construction (Section 6.4).
2. A *mechanism-level finding* (pre-registered family A, robust on 3/3 datasets): within a shared pipeline, soft retention improves Tail-F1 over hard deletion under high asymmetric noise, with no detectable aggregate cost and consistently positive macro means; the direction survives real labelling errors and a 100k-window scale check (Section 6.2).
3. *Boundary results reported as first-class findings*: the graph diagnostic adds no detectable increment over Confident Learning (18/18 Holm-corrected tests null; Tail-F1 effect ceiling +0.026), and the evidence-rescue weighting form adds nothing over a simple frequency floor—on one dataset it is significantly worse (Section 6.3).
4. *Dataset-native corroboration*: applying published corrected labels to CICIDS shows that ∼95% of its rare "Web Attack – Brute Force" class are labelling artefacts, tying the audit directly to real, non-synthetic labelling errors (Section 6.4).

We deliberately bound our claims. The graph-based method is not an accuracy advance: under the frozen protocol Confident Learning is the bar, and we make no claim of reproducing or outperforming official CoLD (our comparator is a CoLD-aligned proxy; Section 7.1). The distinctive value of this study is the audit protocol, the retention mechanism, and the boundary conditions—not a state-of-the-art number.

2 Related Work

Noisy-label learning.

Approaches divide into robust-training and dataset-purification families. Robust training modifies losses or training dynamics: loss correction with a transition matrix [4], peer-network small-loss selection (Co-teaching [5], Co-teaching+ [6]), disagreement-based updates (Decoupling [7]), and semi-supervised treatment of suspected-noisy data (DivideMix [8]). Purification detects and removes mislabeled instances: confident learning [9], representation-based filtering (FINE [14]), and, in security, MORSE [15] and MCRe [16]. CoLD [1] is the closest prior work; we evaluate against a *CoLD-aligned proxy* and are explicit about what it does and does not implement (Section 7.1). Our departure is twofold: we keep the diagnostic in label space (avoiding the distance-metric fragility CoLD itself criticizes), and we replace deletion with signal-preserving weighting motivated by the semi-supervised insight that suspected-noisy samples carry usable signal [8, 15].

Benchmark label errors and dataset audits.

A parallel literature documents that the benchmarks themselves are flawed. Engelen et al. [11] diagnosed labelling and generation errors in CICIDS-2017; Liu et al. [12] reverse-engineered and corrected its labelling logic; Lanvin et al. [13] showed that such errors materially change measured detector performance. We build on this line in two ways: we use the published corrected labels as a source of *real* labelling errors for validation, and we extend the error taxonomy from label mistakes to *label-construction proxy leakage*—features and split structures that let a model reconstruct the labelling rules—with an isolation protocol and regression tests that other studies can reuse.

Graph learning for security telemetry.

Security events are relational, and graph neural networks [17–20] naturally propagate contextual signals. Provenance-graph IDS such as Flash [22] and Argus [23]

demonstrate the value of structure for enterprise detection. We use graph representation learning as a backbone but, unlike embedding-distance denoisers, keep the noise diagnostic in label space so it stays aligned with the classifier.

SOC alert handling and concept drift.

Operational systems must also cope with alert volume and distribution shift. Alert-classification and triage work [25, 26] and drift-aware detection [24] motivate our operational framing, though our contribution is denoising for rare-class signal retention rather than a deployed triage study.

Contrastive self-supervision.

Our representation stage uses cross-view contrastive learning [21], adapted so that the same node in two views forms a positive pair.

2.1 Positioning and Open Challenges

Noisy-label learning for intrusion detection

Learning from noisy labels has become an important challenge in security analytics because network telemetry is often collected through heterogeneous sources, automated rules, and delayed analyst feedback. Unlike conventional computer vision benchmarks where label corruption can often be controlled synthetically, security labels may reflect incomplete attack knowledge, imbalanced detection rules, and evolving adversarial behaviors [30, 31]. Therefore, improving robustness requires not only higher classification accuracy but also understanding which samples should be trusted and which samples contain useful but uncertain training signal.

Existing noisy-label learning methods can be broadly divided into robust training and dataset purification. Robust training approaches attempt to reduce the influence of corrupted samples during optimization. Loss correction methods estimate label transition characteristics and modify the training objective [32]. Sample-selection

approaches exploit uncertainty, loss dynamics, or disagreement between models to identify potentially noisy instances [38, 39]. Co-teaching and related disagreement-based approaches train multiple networks and exchange selected small-loss samples to reduce memorization of corrupted labels [5, 6]. Decoupling further separates the decision of when to update from how to update, demonstrating that disagreement between classifiers can provide useful information under label corruption [7].

A different family focuses on dataset purification. Confident Learning estimates label uncertainty by identifying statistically inconsistent annotations [9]. FINE explores representation-space filtering of noisy instances through latent feature structures [14]. In security domains, MCRe and MORSE extend noisy-label handling toward malicious traffic and malware classification scenarios [15, 16]. More recently, intrusion detection studies have explored contrastive learning and representation enhancement as a way to improve robustness against noisy or incomplete observations [33, 34].

CoLD provides the closest foundation for our work by introducing collaborative label denoising specifically for network intrusion detection. However, existing purification-oriented methods generally treat suspicious samples as removable noise. This assumption is problematic for imbalanced classification because rare-class samples may simultaneously appear inconsistent and be important to the learned decision boundary. Graph-CoLD therefore does not aim to replace purification, but instead explores whether structural context and signal-preserving weighting can retain informative suspicious instances.

Graph learning and contrastive representation for security

Graph representation learning has become increasingly important for security analytics because network activities naturally form relational structures among hosts, addresses, sessions, and temporal events. Graph convolutional networks and graph attention networks demonstrated that neighborhood information can improve representation learning when samples are connected through meaningful relations [17, 18].

Inductive graph representation learning further enables scalable modeling of previously unseen nodes [19], while heterogeneous graph transformers extend graph reasoning to multiple relation types [20].

In security applications, graph methods have mainly focused on attack detection, provenance analysis, and contextual behavior modeling. Systems such as Flash demonstrate the value of provenance graph representation learning for enterprise intrusion detection [22]. Other work investigates security-oriented representation learning and graph analytics for detecting suspicious behaviors [23]. These approaches usually focus on discovering malicious structures from graph embeddings.

Graph-CoLD differs from these methods in its objective. The graph structure is not directly used as an anomaly detector; instead, it provides contextual signals for a label-space consistency diagnostic. This distinction is important because embedding distance alone may capture feature similarity while remaining disconnected from the classifier's final prediction behavior. By combining graph context with prediction-space disagreement, Graph-CoLD aims to identify unreliable labels while preserving samples that remain informative.

SOC alert triage, prioritization, and concept drift

Modern SOC environments face not only detection accuracy challenges but also large-scale alert management problems. Excessive alerts, evolving attack behaviors, and changing network distributions create operational pressure for analysts. Alert classification and active-learning approaches have explored ways to reduce analyst workload by improving alert categorization [25]. Reinforcement-learning-based alert prioritization methods further investigate how automated systems can rank alerts according to operational importance [26].

Beyond static classification, concept drift has become a critical challenge in long-running security monitoring. Detection systems must adapt when normal traffic patterns, attack strategies, and service behaviors evolve. Studies such as CADE,

RecDA, and Insomnia investigate drift detection and adaptation mechanisms for network intrusion detection [24, 40, 41]. Recent SOC-oriented research also explores alert aggregation and filtering through deep learning models to improve investigation efficiency [35–37].

These operational studies motivate our security framing, but Graph-CoLD addresses a different layer of the problem. Rather than replacing analyst triage or predicting attack priority directly, we focus on improving the quality of the training data that supports downstream detection. The key difference is that Graph-CoLD combines three principles: label-space consistency diagnosis, signal-preserving retention, and leakage-audited evaluation. This design aims to answer when graph-based denoising genuinely helps security analytics and when structural information is insufficient.

3 Problem Formulation

Let V be the set of training flows and $\mathcal{M}$ the set of *active* graph views. Each view $m \in \mathcal{M}$ induces a graph $G^m = (V, E^m)$ over the same nodes. The observed training label $\tilde{y}_v$ may differ from the unobserved clean label y_v. We seek a classifier that (i) resists the effect of noisy labels, and (ii) retains useful training signal—especially clean samples of rare classes that a purifier is tempted to discard.

We adopt three design principles. First, the noise diagnostic operates in *label/prediction space*, not embedding distance, so it remains aligned with the classifier and does not re-introduce the local-consistency fragility of distance-based purifiers. Second, denoising is *signal preserving*: a suspicious but informative sample is down-weighted rather than removed. Third, the evaluation is traceable to audited real data, with explicit scope limits.

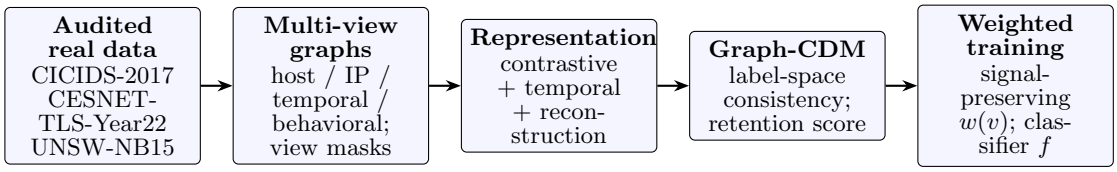


Fig. 1 Graph-CoLD pipeline. Only views supported by a dataset's verified schema are activated; unsupported views are disabled rather than fabricated. The consistency diagnostic (Graph-CDM) is computed in label space and feeds a signal-preserving soft weight and a downstream classifier. Datasets are CICIDS-2017, CESNET-TLS-Year22, and UNSW-NB15 (active views vary by dataset; see Table 1).

4 Method

Graph-CoLD is a two-stage extension of CoLD: self-supervised multi-view representation learning, followed by a label-space consistency diagnostic that drives a signal-preserving weight (Figure 1).

4.1 Multi-view graph construction

Views are built from semantically related feature blocks over the training nodes: a *host* view links samples sharing endpoint identifiers; an *IP* view uses communication/flow statistics, ports, protocol and TLS metadata; a *temporal* view links samples within a time window; a *behavioral* view groups service, protocol, port-state, and related behavioral feature blocks (it is a feature-block view, not host-process lineage) and is used only where such fields exist. Support is part of the reproducible dataset contract: CICIDS-2017 activates host/IP/temporal, CESNET-TLS-Year22 activates IP/temporal, and UNSW-NB15 activates temporal/behavioral. Threat-intelligence views are disabled when the fields are absent. Crucially, edges never cross the train/test split, and (Section 6.4) exact/near-duplicate edges are removed.

4.2 Self-supervised representation

Each active view produces a node embedding z_v^m from Bernoulli-masked input features, $x'_v = x_v \odot \text{Bernoulli}(1-\delta)$. Cross-view contrastive learning treats the same node in

two views as a positive pair and other batch nodes as negatives:

$$\mathcal{L}_{\text{con}} = -\log \frac{\exp(\text{sim}(z_v^a, z_v^b)/\tau)}{\sum_u \exp(\text{sim}(z_v, z_u)/\tau)}. \tag{1}$$

Temporal alignment and reconstruction terms regularize the embedding, $\mathcal{L}_{\text{rep}} = \mathcal{L}_{\text{con}} + \alpha\mathcal{L}_{\text{temporal}} + \beta\mathcal{L}_{\text{recon}}$; active views are fused by mean aggregation to obtain z_v.

4.3 Graph-CDM: a label-space consistency diagnostic

For each node v,

$$\text{GraphCDM}(v) = \lambda_1 D_{\text{pred}}(v) + \lambda_2 D_{\text{neigh}}(v) + \lambda_3 D_{\text{view}}(v), \tag{2}$$

where the prediction term compares per-view predicted labels with the observed *noisy* training label,

$$D_{\text{pred}}(v) = \frac{1}{M} \sum_{m=1}^{M} \mathbf{1}[\tilde{y}_v^m \neq \tilde{y}_v], \tag{3}$$

the neighborhood term is a label-space KL divergence between a node's soft prediction and its neighborhood mean, $D_{\text{neigh}}(v) = \text{KL}\big(\hat{y}_v \,\|\, |N(v)|^{-1} \sum_{u \in N(v)} \hat{y}_u\big)$, and the view term measures active-view mode disagreement, $D_{\text{view}}(v) = 1 - \max_c \frac{1}{M} \sum_m \mathbf{1}[\tilde{y}_v^m = c]$. A fourth provenance-chain term explored in early drafts is removed (frozen λ_4=0): none of the three datasets carries verified provenance chains, and no reported result depends on it. No term uses clean labels or the noise mask; this is enforced by a regression test.

4.4 Signal-preserving soft weighting

The internal evidence score protects samples that would be costly to discard, $e(v) = \text{freq_protect}(\tilde{y}_v)\,(1 + \gamma\,\text{anom}(v))$, min-max normalized to $\tilde{e}(v)$. Rather than a small additive floor (which collapses to hard deletion), we use an evidence-*rescue* form so

that suspicious yet high-evidence samples retain real weight:

$$w(v) = \underbrace{\sigma\big(-\kappa(\mathrm{GraphCDM}(v) - \theta)\big)}_{\text{clean branch}} + \underbrace{\big(1 - \sigma(\cdot)\big)\,\lambda_r\,\tilde{e}(v)}_{\text{evidence rescue}}. \tag{4}$$

The classifier minimizes $\sum_v w(v)\,\mathrm{CE}(f(z_v), \tilde{y}_v)$. Setting $\lambda_r = 0$ and thresholding recovers hard deletion; this is our `ablation_hard` comparator, sharing the identical graph, representation, and diagnostic so that the ablation isolates *only* the weighting. Two further frozen comparators isolate whether the specific *rescue form* (rather than retention per se) carries the benefit: a frequency-floor reweighting $w(v) = \max(\tau_{\mathrm{ret}}, \mathrm{freq}(v))$ with $\mathrm{freq}(v) = \min(1, \text{median class count}/\mathrm{count}(\tilde{y}_v))$, and a no-graph class-balanced cross entropy with effective-number weights ($w_c \propto (1-\beta)/(1-\beta^{n_c})$, β=0.9999) [10]; both share the encoder and training budget (pre-registered family C, Section 6.2).

We stress the intended interpretation of this internal score: "evidence" here means a heuristic estimate of training-signal retention value, not SOC or forensic evidence. Frequency protection encodes *deletion cost*, not trustworthiness. A rare sample is protected because discarding it is costly—it may be one of few training examples for a tail class—not because rarity implies a clean label. The score is thus a heuristic proxy for a risk-aware retention signal $R_{\mathrm{delete}}(v) \propto \mathrm{Cost}(\tilde{y}_v)\,\big(1+\gamma\,\mathrm{anom}(v)\big)$, where Cost is high for rare classes, and the weighting in Eq. (4) implements risk-aware sample retention: samples with high diagnostic suspicion but high deletion cost keep reduced—never zero—influence. A full decision-theoretic treatment that models $P(\text{clean} \mid x) \cdot \mathrm{Cost}(y)$ explicitly is left to future work; the empirical ordering in Section 6.2, where confident-learning-precision pruning outperforms our heuristic, indicates the value of sharpening the $P(\text{clean} \mid x)$ component. A structural limitation is that freq_protect is keyed on the *observed* noisy label $\tilde{y}_v$: under asymmetric tail-to-majority noise, a flipped tail

sample is observed as the majority class and receives weaker protection, while a majority sample mislabelled into a tail class is over-protected. Consistent with this, the pre-registered family C comparison (Section 6.2) finds no detectable advantage of the rescue form over the simple frequency floor—retention itself, not the weighting form, carries the benefit. Both the CoLD-aligned proxy baseline and the `ablation_hard` variant use the same shared encoder architecture, training budget, and view-masking configuration as Graph-CoLD; they differ only in the presence or absence of graph views and the weighting rule applied after Graph-CDM scoring.

4.5 Graph-CoLD Optimization Procedure

The complete Graph-CoLD pipeline consists of representation learning, label-space consistency estimation, retention scoring, and weighted classifier optimization. Algorithm 1 summarizes the complete procedure.

Algorithm 1: Graph-CoLD signal-preserving noisy-label learning.

Input: Multi-view graphs $\{G_m\}_{m=1}^{M}$, noisy labels $\tilde{Y}$, feature matrix X.

Output: Trained classifier f.

1. Construct active graph views according to available telemetry fields.
2. Learn view-specific representations using contrastive, temporal-alignment, and reconstruction objectives, then fuse them into node embeddings Z.
3. Generate prediction-space and neighborhood consistency statistics and compute a Graph-CDM score for each training sample.
4. Estimate internal retention scores from label frequency and anomaly factors, then compute the signal-preserving weights $w(v)$.
5. Optimize the classifier using weighted cross entropy and return f.

The design follows three considerations. First, the diagnostic operates in label and prediction space rather than pure embedding distance. This maintains consistency with the classifier decision process and avoids relying solely on representation similarity. Second, the weighting mechanism adopts an signal-rescue formulation because

suspicious samples may still contain useful class information. A simple lower bound would approximate hard deletion, whereas the rescue formulation allows uncertain but informative samples to participate in training. Third, hard deletion is retained as the primary ablation because it isolates the contribution of signal preservation while keeping the same graph representation and diagnostic components.

4.6 Frozen configuration and tuning discipline

All experiments use one frozen configuration, fixed before the final experiment grid and archived with the reproduction package: diagnostic weights λ_{pred}=0.4, λ_{neigh}=0.3, λ_{view}=0.3 (λ_4=0, removed); weighting κ=4.0, θ=0.5, λ_r=0.8, γ=1.0, log-scaled freq_protect, retention floor τ_{ret}=0.1; hard-deletion threshold CDM $\leq$ 0.5 on the shared diagnostic. Graph construction: host/IP views use k=10 Euclidean nearest neighbours, the temporal view links k=2 neighbours within a 3,600 s window; the encoder is a 2-layer, 4-head heterogeneous graph transformer (hidden 128, dropout 0.2), trained 100 epochs (stage 1) plus 50 (stage 2), learning rate 10^{-3}, batch 128. *Tuning discipline:* no clean validation set and no true injected noise rate is available to any method at any point; hyperparameter selection uses defaults or criteria computed on the noisy training set only (e.g. cleanlab-estimated noise rates for the Co-Teaching forget schedule). This discipline is part of the frozen protocol and applies identically to every comparator.

5 Experimental Design

5.1 Datasets and protocol

Table 1 summarizes three benchmark datasets with different tasks. CICIDS-2017 and UNSW-NB15 are intrusion-detection datasets. CESNET-TLS-Year22 [42] is a TLS service-classification dataset whose SNI-derived application labels do not carry attack/benign semantics; we retain it only as a long-tail service-class stress test and do

Table 1 Datasets and active graph views. Source: `tables/table_1_dataset_protocol.csv`.

Dataset	Classes	Sample policy	Active views
CICIDS-2017	11	post-filter, exact-dedup	host/IP/temporal
CESNET-TLS-Year22	25	audit window	IP/temporal
UNSW-NB15	9	partition (no Worms)	temporal/behavioral

not use it to support attack-level claims. CICIDS-2017 is used under an 11-class *post-filter* policy with exact deduplication (24% of raw training rows removed; Section 6.4). CESNET-TLS-Year22 is a year-spanning backbone TLS dataset used under a 25-service-class audit-window policy (it replaces MALTLS-22, for which we lacked a verified license path). UNSW-NB15 is used under its partition with the smallest class (Worms) removed. Noise semantics per dataset: the asymmetric flip target is benign for the two IDS datasets and a high-frequency service class for CESNET; legacy result filenames may use `tail-to-benign` as an implementation label, but that term has no semantic meaning for CESNET.

Frozen primary protocol.

All headline results use a *frozen final protocol*, fixed before the final experiment grid; any later deviation is disclosed as a protocol amendment (Appendix). The protocol has two ingredients per dataset. (i) *Proxy-field masking*: features that participate in each dataset's label construction, or are strong proxies of it, are removed before any graph or model sees the data (CICIDS: tier-1 mask; CESNET and UNSW: tier-2, which additionally removes the `CATEGORY` one-hot columns—parents of the CESNET label—and the `ct_*` generated features of UNSW). (ii) *Strongest-available group-disjoint split*: CICIDS uses a temporal split ordered by source-file day (with exact training-row deduplication); CESNET uses a strict temporal boundary with zero train–test entity overlap (S1∩S2); UNSW uses its official fixed partition, whose released files lack timestamp and entity columns (a limitation we measure rather than hide: the official partition itself contains 38.4% exact train–test feature collisions after masking).

Split taxonomy and audit measurements.

The audit distinguishes S0 (iid stratified), S1 (temporal), S2 (entity-disjoint, grouped so that no entity appears on both sides), and S3 (leave-one-scenario-out, defined only where the scenario is not identical to the class label: attack-day holdout for CICIDS; undefined for UNSW and CESNET, where S1∩S2 substitutes). For every split we measure and report the train–test exact and near-duplicate intersection (cosine similarity ≥ 0.999 on standardized features). Test-time inference is inductive (per-sample); a transductive variant that builds graphs over test batches is reported only as a control and does not represent an online IDS. On CICIDS, entity isolation is infeasible by construction—the flow graph collapses into one giant connected component spanning 99.96% of valid flows—which we document as a dataset-level finding (Section 9).

Noise injection.

Noise is injected into training labels only; test labels stay clean. The rare-class analysis uses asymmetric noise concentrated on tail classes (training class count $\leq$ window median) at rates $\{40, 60, 80\}\%$ with seeds 0–9 bound to the injection. Flip targets are the benign class for the two IDS datasets and the high-frequency service class `microsoft-diagnostic` for CESNET, which has no benign semantics. Symmetric and graph-consistency noise were explored only under the legacy protocol and are not part of the formal result matrix. The complete mapping matrix ships with the reproduction package (`noise_map.json`).

Pre-registered statistical protocol.

The inference unit is the per-noise-rate paired seed comparison (n=10); we do not pool the 30 seed–rate observations as independent. Holm correction is applied within three pre-registered families: family A (9 tests; soft retention vs. hard deletion, Tail-F1), family B (18 tests; CL+Graph-CDM vs. CL, Macro- and Tail-F1—the G2 decision gate), and family C (18 tests; soft weight vs. simple reweighting baselines, Tail-F1).

"No aggregate cost" claims are replaced by TOST non-inferiority at δ=0.01 Macro-F1; because a 0.01 macro change could hide a concentrated single-class regression, a per-class guard additionally requires no class-level F1 regression above 0.05. Seed random-intercept mixed-effects models provide robustness confirmation. Success requires Holm $p < 0.05$, $d_z \geq 0.3$, and a bootstrap 95% CI excluding zero; the verdict vocabulary is frozen (*robust*: $\geq$2 datasets pass; *narrow*: 1; *null*: 0).

Windows and scale.

Each dataset uses a deterministic audit window of 10,000 training and 5,000 test instances drawn inside the frozen split. A one-shot scale validation on CICIDS (train 100,000 / test 20,000; 3 seeds $\times$ 3 rates) checks that conclusions do not flip with scale; they do not (Section 6.2). We are explicit that this is a focused audit study, not a full-archive evaluation.

5.2 Comparators and metrics

Within the frozen protocol we compare six configurations that share one encoder architecture and training budget: Graph-CoLD-soft; hard deletion (`ablation_hard`); the frequency-floor and class-balanced-CE comparators of Section 4; Confident Learning (cleanlab `find_label_issues` with *full stratified out-of-fold* probabilities, per its sample-out protocol, and a shared-budget classifier head) [9]; and the CL+Graph-CDM combination that soft-reweights CL-retained samples (issue-set intersection/union fusions are exploratory). The rows labelled *CoLD-aligned proxy* were called "CoLD" in an earlier draft; the implementation predates the official CoLD code release and does not implement its multi-subset encoding, clustering-consistency purification, or XGBoost head. We therefore make no claim of reproducing or outperforming official CoLD, whose published numbers arise under a different nine-class protocol (Section 7.1). FINE [14] and a fully-implemented Co-Teaching [5] (forget schedule driven by an *estimated*, not oracle, noise rate) were run under the legacy

stratified audit window and are reported separately as explicitly non-comparable (Section 6.2). Metrics: aggregate Macro-F1, tail-class Macro-F1 (tail = classes with training count $\leq$ window median, fixed per data contract), tail recall, and paired deltas with bootstrap 95% CIs under the statistical protocol above.

6 Results

6.1 The frozen protocol reorders the headline comparison

Table 2 reports the full frozen-protocol grid. Two observations frame everything that follows. First, absolute numbers are far below the legacy stratified-window results that opened this project (CICIDS Macro-F1 drops to $\approx$0.14): proxy masking and group-disjoint splits remove the duplicate memorization and label-construction shortcuts that inflated earlier figures (Section 6.4). Second, every *retention-style* configuration—soft weighting, frequency-floor, class-balanced CE, Confident Learning, and the CL+Graph combination—lands in one narrow band per dataset, while hard deletion falls off it on tail classes (CICIDS Tail-F1 0.097 vs. 0.355–0.358 for the retention band; UNSW 0.161 vs. 0.200–0.263). No single method "wins" this table, and we draw no headline ranking from it; the meaningful contrasts are the three pre-registered hypothesis families below. FINE and Co-Teaching, which were run under the legacy stratified window, are reported separately in Table 3 and are not comparable to this grid.

6.2 RQ-A: Soft retention preserves tail signal; the specific weighting form does not matter

Family A (robust).

Table 4 reports the pre-registered soft-vs.-hard comparison under the frozen protocol. All nine conditions pass: Holm-corrected $p < 0.05$, effect sizes d_z from 0.77 to 3.14, and

Table 2 Final frozen protocol (proxy-masked features, group-disjoint primary splits): mean±std over 10 seeds × asymmetric tail-noise rates {0.4, 0.6, 0.8}. The historical CoLD-aligned proxy is not part of this grid (Section 7.1).

Dataset	Method	Macro-F1	Tail-F1
CICIDS	Graph-CoLD-soft	0.136±0.099	0.358±0.322
	Hard deletion (ablation)	0.079±0.083	0.097±0.236
	Freq-floor reweighting	0.140±0.108	0.355±0.325
	Class-balanced CE (no graph)	0.132±0.089	0.356±0.326
	Confident Learning	0.137±0.131	0.355±0.413
	CL + Graph-CDM soft	0.134±0.126	0.342±0.397
CESNET	Graph-CoLD-soft	0.436±0.057	0.249±0.112
	Hard deletion (ablation)	0.407±0.060	0.205±0.118
	Freq-floor reweighting	0.435±0.058	0.249±0.113
	Class-balanced CE (no graph)	0.435±0.058	0.246±0.115
	Confident Learning	0.393±0.032	0.295±0.076
	CL + Graph-CDM soft	0.396±0.034	0.300±0.077
UNSW	Graph-CoLD-soft	0.482±0.029	0.200±0.047
	Hard deletion (ablation)	0.460±0.036	0.161±0.058
	Freq-floor reweighting	0.492±0.032	0.217±0.052
	Class-balanced CE (no graph)	0.488±0.031	0.210±0.052
	Confident Learning	0.485±0.024	0.263±0.055
	CL + Graph-CDM soft	0.483±0.023	0.256±0.053

Table 3 Additional noisy-label baselines under the *legacy stratified audit window* (iid, unmasked; the protocol later shown to be inflated by duplicate and proxy leakage, Section 6.4). Reported for completeness with estimated (not oracle) noise rates; **not comparable** to the frozen-protocol numbers in Table 2.

Dataset	Method	Macro-F1	Tail-F1
CICIDS	FINE (est. rate)	0.700±0.085	0.403±0.190
	Co-Teaching (est. rate)	0.540±0.040	0.067±0.088
CESNET	FINE (est. rate)	0.695±0.093	0.501±0.180
	Co-Teaching (est. rate)	0.495±0.122	0.159±0.224
UNSW	FINE (est. rate)	0.461±0.029	0.160±0.069
	Co-Teaching (est. rate)	0.359±0.006	0.000±0.000

bootstrap CI lower bounds above zero on every dataset and rate. The effect is large on CICIDS (+0.21 to +0.31 Tail-F1) and small-to-moderate on CESNET/UNSW (+0.03 to +0.05). Aggregate cost is bounded on three sides: TOST non-inferiority at δ=0.01 passes 9/9, the per-class guard finds no class regression beyond −0.036 (threshold 0.05), and paired macro means are themselves positive (+0.014 to +0.062)—so we

Table 4 Family A (pre-registered, 9 Holm-corrected tests): soft retention vs. hard deletion, Tail-F1, final frozen protocol. Macro Δ is the paired macro-F1 mean difference; worst class is the largest per-class F1 regression (guard ≤0.05). Verdict: **robust** (3/3 datasets).

Dataset	Rate	Δ Tail-F1	Holm p	d_z	95% CI	Macro Δ / worst class
CICIDS	0.4	+0.314	0.017	1.00	[+0.134, +0.498]	+0.052 / +0.000
	0.6	+0.259	0.024	0.86	[+0.081, +0.433]	+0.062 / +0.000
	0.8	+0.209	0.024	0.77	[+0.050, +0.371]	+0.055 / +0.000
CESNET	0.4	+0.034	0.015	1.08	[+0.017, +0.053]	+0.024 / -0.029
	0.6	+0.052	<0.001	2.82	[+0.042, +0.063]	+0.031 / -0.036
	0.8	+0.047	<0.001	3.14	[+0.039, +0.056]	+0.031 / -0.008
UNSW	0.4	+0.025	0.009	1.23	[+0.012, +0.036]	+0.014 / -0.005
	0.6	+0.040	<0.001	2.76	[+0.032, +0.049]	+0.023 / -0.000
	0.8	+0.050	<0.001	2.31	[+0.038, +0.064]	+0.030 / -0.001

state "no detectable aggregate cost, with consistently positive macro means," not "no cost." A seed random-intercept mixed-effects model confirms the result on each dataset (mean Δ Tail-F1 +0.044/+0.261/+0.039 for CESNET/CICIDS/UNSW, one-sided $p < 10^{-4}$). Mechanistically, hard deletion removes clean rare-class training rows flagged as suspicious; soft weighting keeps them at reduced influence.

The direction survives real labelling errors and scale.

Replacing synthetic injection with the real labelling errors of CICIDS-2017 (training on original released labels, evaluating against Engelen-corrected labels [12]; Section 6.4) preserves the direction: soft − hard Tail-F1 = +0.284 [+0.089, +0.487] (Table 5; descriptive, 10 paired seeds). A one-shot scale validation at train 100,000 / test 20,000 on CICIDS also preserves the direction at every rate (+0.150/+0.056/+0.031), while showing that window size itself dominates tail learnability (hard deletion reaches 0.698 Tail-F1 at rate 0.4 in the 100k window vs. 0.093 in the 10k window).

Table 5 Real labelling errors (CICIDS-2017, Engelen-corrected reference; descriptive, 10 seeds, frozen temporal window). Training uses the original released labels (4.7% real train-pool errors); evaluation uses corrected labels. The soft-vs-hard direction holds under real noise: Δ Tail-F1 $+0.284$ $[+0.089, +0.487]$.

Method	Macro-F1	Tail-F1
Graph-CoLD-soft	0.257	0.865
Confident Learning	0.270	0.910
Hard deletion	0.184	0.581
CoLD-aligned proxy	0.050	0.000
Soft − hard, Tail-F1: $+0.284$ $[+0.089, +0.487]$ (paired, n=10)		

Family C (null): the rescue form adds nothing over simple reweighting.

Table 6 tests whether the specific evidence-rescue form—the design element that distinguishes our weighting from trivial baselines—beats a frequency floor or class-balanced CE. It does not: no condition survives Holm correction in the positive direction, and on UNSW the soft weight is significantly *worse* than the frequency floor at rates 0.4/0.6 (CI upper bounds $-0.011/-0.013$). We therefore withdraw the claim that evidence-rescue weighting is superior to simple reweighting. The surviving mechanism claim is deliberately narrow: *retention* beats *deletion* (family A); among retention schemes, the weighting form is empirically interchangeable, and a practitioner can obtain the benefit with a frequency floor alone.

We retain an honesty note from earlier drafts: a "rare-evidence recovery" diagnostic that counted a sample as recovered whenever it was retained was tautological and near-ceiling by construction; it was removed from the hypothesis families, and all headline comparisons rely on held-out tail-F1 only.

6.3 RQ-B: When graph structure does not help

This section collects the negative results as first-class findings.

Table 6 Family C (pre-registered, 18 Holm-corrected tests): evidence-rescue soft weight (#4) vs. simple reweighting baselines, Tail-F1. Verdict: **null**; on UNSW the soft weight is significantly *worse* than freq-floor at rates 0.4/0.6 (CI upper bounds −0.011/−0.013). The superiority claim is withdrawn.

Dataset	Rate	Comparison	Δ Tail-F1	95% CI
CICIDS	0.4	vs. freq-floor	+0.0064	[-0.025, +0.039]
	0.4	vs. CBCE	+0.0090	[-0.009, +0.031]
	0.6	vs. freq-floor	-0.0021	[-0.008, +0.002]
	0.6	vs. CBCE	+0.0009	[-0.007, +0.011]
	0.8	vs. freq-floor	+0.0033	[-0.001, +0.008]
	0.8	vs. CBCE	-0.0031	[-0.008, +0.000]
CESNET	0.4	vs. freq-floor	-0.0040	[-0.010, +0.002]
	0.4	vs. CBCE	-0.0034	[-0.012, +0.005]
	0.6	vs. freq-floor	+0.0065	[+0.001, +0.012]
	0.6	vs. CBCE	+0.0105	[+0.004, +0.016]
	0.8	vs. freq-floor	-0.0017	[-0.007, +0.004]
	0.8	vs. CBCE	+0.0040	[-0.001, +0.010]
UNSW	0.4	vs. freq-floor	-0.0194	[-0.027, -0.011]
	0.4	vs. CBCE	-0.0102	[-0.022, +0.000]
	0.6	vs. freq-floor	-0.0216	[-0.031, -0.013]
	0.6	vs. CBCE	-0.0139	[-0.025, -0.003]
	0.8	vs. freq-floor	-0.0102	[-0.020, +0.001]
	0.8	vs. CBCE	-0.0062	[-0.019, +0.003]

Family B (null): no detectable increment over Confident Learning.

The pre-registered decision gate for the graph diagnostic asked whether soft-reweighting the samples that Confident Learning retains adds anything over CL alone. Table 7 reports the complete 18-test family: no test is significantly positive after Holm correction (all $p \geq 0.608$), and the largest Tail-F1 CI upper bound is +0.026 (CESNET, rate 0.6)—an informative effect ceiling, not merely an absence of evidence. Mixed-effects robustness models agree (per-dataset mean Δ Tail-F1 +0.004/−0.014/−0.007, none supporting an increment). We accordingly withdraw the claim that Graph-CDM adds increment over precise probabilistic pruning and reposition it as an audit/diagnostic instrument.

Table 7 Family B (pre-registered, complete 18 Holm-corrected tests, no placeholders): CL+Graph-CDM vs. Confident Learning alone. No test is significantly positive; decision gate G2 = **null**. Largest Tail-F1 CI upper bound: +0.026 (CESNET, 0.6).

Dataset	Rate	Metric	Δ	Holm p	95% CI
CICIDS	0.4	Macro	+0.0115	1.000	[-0.013, +0.038]
	0.4	Tail	+0.0040	1.000	[+0.000, +0.012]
	0.6	Macro	-0.0230	1.000	[-0.065, +0.003]
	0.6	Tail	-0.0422	1.000	[-0.127, +0.000]
	0.8	Macro	+0.0016	1.000	[-0.005, +0.011]
	0.8	Tail	-0.0032	1.000	[-0.010, +0.000]
CESNET	0.4	Macro	+0.0023	1.000	[-0.004, +0.009]
	0.4	Tail	+0.0038	1.000	[-0.010, +0.020]
	0.6	Macro	+0.0072	0.608	[+0.001, +0.013]
	0.6	Tail	+0.0118	1.000	[-0.003, +0.026]
	0.8	Macro	-0.0027	1.000	[-0.010, +0.004]
	0.8	Tail	-0.0024	1.000	[-0.015, +0.010]
UNSW	0.4	Macro	-0.0035	1.000	[-0.008, +0.000]
	0.4	Tail	-0.0102	1.000	[-0.021, -0.000]
	0.6	Macro	-0.0031	1.000	[-0.007, +0.001]
	0.6	Tail	-0.0109	1.000	[-0.022, -0.000]
	0.8	Macro	+0.0006	1.000	[-0.002, +0.004]
	0.8	Tail	+0.0010	1.000	[-0.006, +0.009]

Saturation at scale.

In the 100k-window scale validation, CL alone is near ceiling (Tail-F1 0.95–0.995); the combination has no headroom, and the rate-0.8 point estimate is negative (-0.014 $[-0.042, +0.001]$). Under real labelling errors the ordering is the same: CL reaches 0.910 Tail-F1 vs. 0.864 for Graph-CoLD-soft (Table 5).

Transductive construction adds nothing.

A transductive control that builds graphs over test batches performs *worse* than the inductive classifier on all three datasets in the audit screening (e.g. CICIDS 0.603 vs. 0.684 Macro-F1), ruling out—for this implementation—the concern that test-batch information drives the reported numbers. This is evidence about our pipeline, not a general claim about transductive methods.

Boundary conditions as testable hypotheses.

Where might a graph diagnostic still add value? The informativeness pattern (Section 6.5) and the family-B ceiling suggest two registered hypotheses for future work rather than salvage claims: (i) graphs built from verified relational telemetry (host lineage, authenticated flows) rather than feature-space neighbourhoods; (ii) noise processes correlated with graph structure. Our real-noise experiment does not support the second hypothesis for CICIDS's actual labelling errors, where CL remains ahead.

6.4 RQ-C: Where inflated numbers come from (leakage decomposition)

Historical account.

Our initial pipeline reported a ∼28-point CICIDS advantage with a curve that barely degraded under 60% noise—implausible for a denoiser. The first audit found 152,201/630,399 (≈24%) exact-duplicate CICIDS training rows and clean-label paths in score/graph construction; removing them collapsed the apparent advantage (to +6.7 points on CICIDS under the interim audit protocol). Those interim numbers motivated the frozen protocol and are superseded by the systematic decomposition below.

Four-factor decomposition under the final protocol.

Table 8 flips one leakage factor at a time against the frozen protocol: duplicate records (DUP), cross-split entities (ENT), label-construction proxy fields (PRX), a clean-label oracle path in the diagnostic (ORA), and all at once (ALL). On CESNET, entity overlap and proxy fields contribute +0.187 and +0.184 Tail-F1 each, approximately additive (ALL +0.249 soft; +0.359 for the deletion-style proxy)—a 25–36 point inflation across methods and metrics, reproducing the historical "+28-point" figure in controlled form. On UNSW the picture inverts: the official fixed partition already contains 38.4% exact train–test feature collisions, so adding deduplication

Table 8 Four-factor leakage decomposition (descriptive, 10 seeds, rate 0.6): Tail-F1 Δ of each leakage condition vs. the final frozen protocol. *: bootstrap 95% CI excludes zero. CICIDS ENT is infeasible by construction (giant connected component; Section 9). Leakage flatters deletion-style methods most and compresses the apparent soft-retention advantage.

Dataset	Method	DUP	ENT	PRX	ORA	ALL
CICIDS	CoLD-aligned proxy	+0.467*	n/a	+0.303*	+0.350*	+0.389*
	Hard deletion	+0.375*	n/a	+0.209*	+0.257*	+0.296*
	Soft retention	+0.244	n/a	+0.020	+0.007	+0.041
CESNET	CoLD-aligned proxy	-0.002	+0.209*	+0.129*	+0.098*	+0.359*
	Hard deletion	-0.014	+0.196*	+0.203*	+0.045*	+0.305*
	Soft retention	-0.011	+0.187*	+0.184*	-0.008	+0.249*
UNSW	CoLD-aligned proxy	-0.107*	+0.016*	-0.031*	+0.084*	+0.024*
	Hard deletion	-0.106*	+0.024*	-0.040*	+0.058*	-0.002
	Soft retention	-0.115*	+0.002	-0.052*	-0.007	-0.052*

lowers scores by 0.115 (duplicates inflate the frozen numbers from inside the official split), and unmasking `ct_*` proxies is net negative because feature-space collisions dilute (−0.052).

Leakage flatters deletion, not retention.

The decomposition reverses a natural misreading of our earlier drafts. Leakage factors inflate *deletion-style* methods most (CICIDS CoLD-aligned proxy: DUP +0.467, ALL +0.389; hard deletion +0.375/+0.296), because under the clean protocol they delete tail classes nearly to zero (proxy Tail-F1 0.000 on CICIDS) and any leakage "finds back" those classes for free. Soft retention, having already banked its tail gains, has little headroom (CICIDS DUP +0.244 with CI crossing zero). Consequently, leaky protocols *compress* the soft-vs.-hard gap (CICIDS: 0.256 under the frozen protocol vs. 0.125 under DUP)—they flatter deletion baselines and mask their true tail cost. The oracle path (ORA) likewise helps only deletion-style methods.

Dataset-level findings.

Three findings are properties of the benchmarks themselves and transfer beyond this study: (i) CESNET's released features include `CATEGORY` one-hot columns that are

parents of the label—a direct label proxy; (ii) UNSW's official partition ships with 38.4% exact feature collisions after masking; (iii) CICIDS's flow graph collapses into one giant connected component (99.96% of valid flows), making entity-disjoint splitting infeasible by construction (Section 9).

Real labelling errors corroborate the audit.

Applying the published Engelen relabelling rules [12] to the original CICIDS release yields 128,247 corrected rows (4.53% of valid flows): 65,634 missed portscan rows and 31,339 missed DDoS rows were released as BENIGN, while 95% of the "Web Attack – Brute Force" class (1,434 of 1,507 rows) are attack-orchestration artefacts, leaving only 74 true positives. Rare-attack classes in this benchmark are, to a large extent, artefacts of the labelling pipeline—direct, dataset-native corroboration of the label-construction-proxy concern that motivates our audit protocol (cf. [11, 13]).

6.5 Graph informativeness (descriptive, hypothesis-generating)

A per-dataset neighborhood-homophily / view-coverage diagnostic, computed under the legacy protocol, is directionally consistent with where graph-based processing helped most in that era: CICIDS (rich host/IP/temporal views) showed the largest margins, CESNET sat near a performance ceiling, and UNSW (temporal/behavioral views only) the smallest (Figure 2). With only three datasets this pattern is descriptive and hypothesis-generating, not inferential, and under the frozen protocol the family-B result shows that whatever signal the views carry does not translate into an increment over precise pruning. We keep the diagnostic because it is cheap to compute and, as an *audit* instrument, it flags upfront when a dataset's views are too weak for graph construction to be worth the cost.

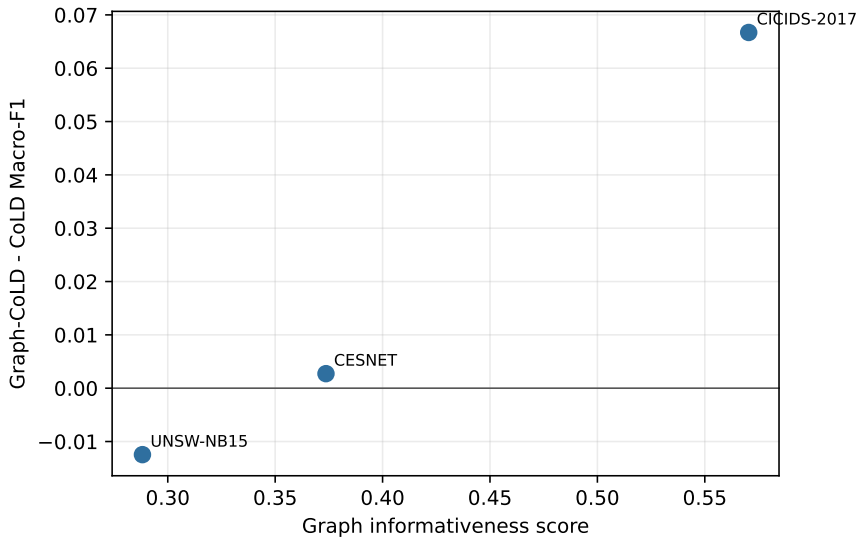


Fig. 2 Graph informativeness vs. margin of graph-based processing under the legacy protocol. Higher neighborhood homophily and view coverage are directionally consistent with larger margins (three datasets; descriptive, hypothesis-generating only).

7 Discussion

The operational motivation is strongest in settings where multiple telemetry sources provide partially consistent signals, such as host relationships, communication patterns, and temporal behavior. In those settings, a suspicious sample need not be useless: a rare attack instance may appear inconsistent because it represents an emerging technique rather than an incorrect label. Our benchmark study tests only this training-data mechanism; it does not establish deployment suitability.

However, the results also demonstrate clear boundaries. When available views are weak or incomplete, graph structure provides limited additional information. The UNSW-NB15 results illustrate this condition: without sufficiently informative relational views, graph-based diagnostics cannot reliably create additional training signal. Similarly, datasets operating near a performance ceiling, such as CESNET-TLS-Year22 under the evaluated protocol, naturally limit the observable improvement.

7.1 CoLD fidelity: what the proxy is and is not

The comparator historically labelled "CoLD" in earlier drafts is a *CoLD-aligned proxy*: a neighbourhood consistency diagnostic with hard thresholding and a shared classifier head, built before the official CoLD code release. It does not implement official CoLD's multi-subset feature encoding, clustering-consistency purification, or XGBoost pipeline. The official NDSS paper reports Macro-F1 in the 0.94–1.00 range on a nine-class refined CICIDS-2017 under random 8:2 splits—a protocol that differs from our 11-class, tail-targeted, group-disjoint audit windows in labelling space, split design, and noise model, so the numbers are not comparable in either direction. We use the proxy solely as a deletion-style anchor inside our frozen protocol and make no claim of reproducing or outperforming official CoLD [1]. An exact-alignment grid under the official protocol is registered as future work.

A key methodological decision in this work is the avoidance of unfair state-of-the-art comparisons. Confident Learning was therefore run inside the frozen protocol itself (Tables 2 and 7), even though it outperforms the graph-based combination wherever a difference is detectable; that result is part of the empirical record, not an inconvenience. Other purification and robust-training variants (MCRe, MORSE, Decoupling) were not reimplemented under the final leakage-audited protocol, and FINE/Co-Teaching ran only under the legacy window (Table 3). Mixing results obtained under different data-cleaning, duplication-removal, and evaluation settings would create an unreliable comparison, so those methods remain excluded from the formal matrix. A comprehensive comparison with all previous methods requires faithful implementations or official code under a unified evaluation framework.

The frozen-protocol ordering carries a practical lesson for benchmark data curation and a hypothesis for later SOC-facing validation. Confident Learning's precise pruning sets the performance bar wherever a difference is detectable, while the retention-style reweightings form one narrow band (Table 2). The differentiator is not deletion versus

retention but *diagnostic precision*: an imprecise purifier sacrifices scarce rare-class training signal whether it deletes samples or down-weights them insufficiently, and family C shows the specific weighting form matters little once retention is granted. Sharpening the $P(\text{clean} \mid x)$ component—for example by feeding confident-learning-grade probabilities into a retention-aware weighting—is the natural next step, and our family-B result gives the effect ceiling such a combination must beat.

The leakage audit provides a broader methodological lesson for graph-based security machine learning. Graph models can unintentionally exploit duplicated records, cross-split connections, or hidden clean-label information. Such effects may produce impressive but misleading improvements. Therefore, graph security studies should explicitly verify data separation, graph construction constraints, and label independence before reporting performance gains.

Future work should extend the audit protocol toward streaming SOC environments, full-archive evaluation, analyst-in-the-loop validation, and broader cross-domain benchmark comparisons. These extensions are necessary to determine whether signal-preserving retention remains effective when labels arrive continuously and attack distributions evolve over time. The results support a deliberately narrow claim set. Within a shared pipeline, soft retention preserves rare-class training signal that hard deletion discards—robustly across datasets, noise rates, real labelling errors, and window scale. Beyond that, our experiments do *not* support: an accuracy advantage for graph-based denoising (family B is null against Confident Learning), a superiority of the specific rescue weighting (family C is null, with one significant reversal), a ranking or triage advantage, or SOC/forensic effectiveness of benchmark-level sample retention.

8 Threats to Validity

Internal. The evaluation uses deterministic audit windows for tractability; absolute scores are far below full-archive numbers, and the 100k scale check shows window size itself dominates tail learnability. The paired, seed-controlled family comparisons—not the absolute numbers—are the objects of inference. Two protocol deviations occurred and are disclosed as amendments (Appendix A): family B was initially run at full power on only one dataset (remediated by complete reruns), and the soft-retention claim was initially supported only by screening-tier evidence (remediated by the full family-A grid). *Construct.* Injected noise is synthetic; we mitigate this with the real-labelling-error experiment, which preserves the family-A direction. The legacy rare-evidence-recovery diagnostic was tautological and is excluded from all hypothesis families. Tail-class definitions are window-relative and frozen per data contract. *External.* CICIDS-2017, CESNET-TLS-Year22, and UNSW-NB15 are benchmarks, not SOC deployments; no analyst study is claimed. The strongest available split differs per dataset (UNSW's official partition lacks timestamp/entity columns and carries 38.4% exact collisions we measure but cannot remove without leaving the official split). MCRe and MORSE pass a clean-label sanity floor but were not re-run under the frozen protocol; we report caveats rather than possibly unfaithful numbers.

9 Limitations

Five limitations bound the scope. (i) *CICIDS entity isolation is infeasible by construction*: the flow graph over the raw release forms one giant connected component covering 99.96% of valid flows, so the cross-split entity factor cannot be measured on this dataset; we document the topology rather than substitute a weaker entity definition. (ii) *UNSW's official partition* carries 38.4% exact train–test feature collisions after masking; we keep the official split for comparability and measure the collision effect in the leakage decomposition instead of silently repartitioning. (iii) *CESNET*

supports only service-class claims: its SNI-derived labels have no attack semantics, so attack-level conclusions are limited to CICIDS/UNSW. (iv) *The CoLD-aligned proxy is not official CoLD*; no cross-comparison with published CoLD numbers is valid in either direction, and an official-code alignment grid is registered as future work. (v) *TOST non-inferiority at* δ=0.01 bounds only the aggregate: the margin equals a worst-case single-class F1 change of $\sim$0.11 in an 11-class task, which is why the per-class guard accompanies every non-inferiority claim. MALTLS-22 and OpTC are not evaluated (no verified license/provenance path); behavioral and threat-intelligence views are disabled where fields are absent; full-archive and streaming evaluations are future work.

10 Conclusion

We set out to extend collaborative label denoising with graph structure and ended up with a boundary study. Under a frozen, leakage-audited protocol— proxy-masked features, group-disjoint splits, pre-registered hypothesis families—the surviving positive result is mechanism-level and robust: soft retention preserves rare-class training signal that hard deletion discards, across three datasets, three noise rates, real labelling errors, and a 100k-window scale check, at no detectable aggregate cost. The graph diagnostic itself adds no detectable increment over Confident Learning (effect ceiling +0.026 Tail-F1), and the specific rescue weighting adds nothing over a frequency floor—results we report as first-class findings with effect ceilings rather than as absences. The four-factor leakage decomposition explains where the earlier, larger numbers came from: duplicates, cross-split entities, proxy fields, and oracle paths inflate deletion-style baselines most and compress the apparent advantage of retention. More broadly, our results argue that security data curation should shift from indiscriminate purification toward signal-aware retention, in which deletion decisions explicitly weigh the class-level cost of discarding scarce training examples against the

precision of the noise diagnostic—and that graph-based denoising results in security should be accompanied by a leakage audit before any performance claim is made. CESNET supports only a service-class stress-test claim; attack-level conclusions are limited to CICIDS and UNSW. We hope the audit protocol, the regression tests, and the honestly reported boundaries are as useful to the community as any single number.

Appendix A Protocol amendments

The protocol was frozen before the final experiment grid. Two deviations occurred; both are disclosed here in full, with their remediations.

PA-1: selective family-B testing (remediated).

Family B was frozen as 3 datasets × 2 metrics × 3 noise rates = 18 tests. In the initial full-power run, only CESNET was executed at full power; CICIDS and UNSW were stopped after a screening pass (3 seeds, one rate) showed non-positive increments. This futility-stopping rule was *not* pre-registered and constituted selective testing. As remediation, the full-power protocol (10 seeds × 3 rates, each dataset's frozen primary protocol) was executed on the two remaining datasets and the Holm correction recomputed over the complete 18-test family with no placeholders (Table 7). The screening-stop decision is disclosed as a post-hoc choice; the G2 verdict (null) is unchanged under the complete family.

PA-2: claim strength ahead of evidence (remediated).

The soft-retention finding was initially reported as a core claim while supported only by screening-tier evidence (3 seeds, one rate). The frozen protocol requires full inference for headline claims. As remediation, the complete family-A grid (10 seeds × 3 rates × 3 datasets under the frozen primary protocol) was executed and the claim re-adjudicated under the pre-registered Holm family, TOST non-inferiority, per-class guard, and mixed-effects confirmation (Table 4); the verdict is *robust* (3/3 datasets).

Appendix B CESNET-TLS-Year22 class-filter policy

CESNET is evaluated as a TLS service-classification stress test, not as an attack dataset. The repository audit report `reports/cesnet_class_policy_report.json` records 178 raw service labels. The frozen post-filter policy retains the 25 labels in Table A1 because each has at least 1,000 rows in the audited source window; the other 153 labels are excluded rather than merged. The largest retained class is capped at 4,958 rows. Counts below are pre-split source-window counts and are not labels of attack prevalence.

Table A1 Frozen CESNET 178-to-25 service-class filter.

Service label	Count	Service label	Count
apple-itunes	1,666	bitdefender-gravityzone	1,943
dropbox	2,472	eset-edf	1,506
eset-edtd	1,324	facebook-graph	1,989
google-www	1,088	instagram	1,323
kaspersky	1,459	microsoft-authentication	3,083
microsoft-defender	1,326	microsoft-diagnostic	4,958
microsoft-onedrive	3,137	microsoft-push	3,516
microsoft-settings	4,808	microsoft-update	4,958
o2tv	1,989	office365	2,584
outlook	3,517	seznam-media	1,787
skype	3,511	spotify	1,217
teams	3,802	tiktok	1,128
vmware-vcsa	1,033		

References

[1] S. Yang, X. Zheng, J. Li, J. Xu, E. C. H. Ngai, CoLD: Collaborative label denoising framework for network intrusion detection, in: Network and Distributed System Security (NDSS) Symposium, 2026. Available: https://dx.doi.org/10.14722/ndss.2026.231950

[2] D. Arp et al., Dos and don'ts of machine learning in computer security, in: USENIX Security Symposium, 2022, pp. 3971–3988.

[3] H. Song, M. Kim, D. Park, Y. Shin, J.-G. Lee, Learning from noisy labels with deep neural networks: A survey, IEEE TNNLS 34 (11) (2022) 8135–8153.

[4] G. Patrini, A. Rozza, A. K. Menon, R. Nock, L. Qu, Making deep neural networks robust to label noise: A loss correction approach, in: CVPR, 2017, pp. 1944–1952.

[5] B. Han et al., Co-teaching: Robust training of deep neural networks with extremely noisy labels, in: NeurIPS, 2018, pp. 8535–8545.

[6] X. Yu et al., How does disagreement help generalization against label corruption?, in: ICML, 2019, pp. 7164–7173.

[7] E. Malach, S. Shalev-Shwartz, Decoupling "when to update" from "how to update", in: NeurIPS, 2017, pp. 960–970.

[8] J. Li, R. Socher, S. C. H. Hoi, DivideMix: Learning with noisy labels as semi-supervised learning, in: ICLR, 2020.

[9] C. G. Northcutt, L. Jiang, I. L. Chuang, Confident learning: Estimating uncertainty in dataset labels, JAIR 70 (2021) 1373–1411.

[10] Y. Cui, M. Jia, T.-Y. Lin, Y. Song, S. Belongie, Class-balanced loss based on effective number of samples, in: CVPR, 2019, pp. 9268–9277.

[11] G. Engelen, V. Rimmer, W. Joosen, Troubleshooting an intrusion detection dataset: the CICIDS2017 case study, in: IEEE Security and Privacy Workshops (WTMC), 2021, pp. 7–12.

[12] L. Liu, G. Engelen, T. Lynar, D. Essam, W. Joosen, Error prevalence in NIDS datasets: A case study on CIC-IDS-2017 and CSE-CIC-IDS-2018, in: IEEE Conference on Communications and Network Security (CNS), 2022, pp. 254–262.

[13] M. Lanvin, P.-F. Gimenez, Y. Han, F. Majorczyk, L. Mé, E. Totel, Errors in the CICIDS2017 dataset and the significant differences in detection performances it makes, in: Risks and Security of Internet and Systems (CRiSIS 2022), LNCS 13857, Springer, 2023, pp. 18–33.

[14] T. Kim, J. Ko, J. Choi, S.-Y. Yun, FINE samples for learning with noisy labels, in: NeurIPS, 2021, pp. 24137–24149.

[15] X. Wu, W. Guo, J. Yan, B. Coskun, X. Xing, From grim reality to practical solution: Malware classification in real-world noise, in: IEEE S&P, 2023, pp. 2602–2619.

[16] Q. Yuan, G. Gou, Y. Zhu, Y. Zhu, G. Xiong, Y. Wang, MCRe: A unified framework for handling malicious traffic with noise labels, IEEE Transactions on Information Forensics and Security, 2023.

[17] T. N. Kipf, M. Welling, Semi-supervised classification with graph convolutional networks, in: ICLR, 2017.

[18] P. Veličković et al., Graph attention networks, in: ICLR, 2018.

[19] W. L. Hamilton, R. Ying, J. Leskovec, Inductive representation learning on large graphs, in: NeurIPS, 2017.

[20] Z. Hu, Y. Dong, K. Wang, Y. Sun, Heterogeneous graph transformer, in: The Web Conference (WWW), 2020, pp. 2704–2710.

[21] T. Chen, S. Kornblith, M. Norouzi, G. Hinton, A simple framework for contrastive learning of visual representations, in: ICML, 2020, pp. 1597–1607.

[22] M. U. Rehman, H. Ahmadi, W. U. Hassan, Flash: A comprehensive approach to intrusion detection via provenance graph representation learning, in: IEEE S&P,

2024, pp. 3552–3570.

[23] J. Xu, X. Shu, Z. Li, Understanding and bridging the gap between unsupervised network representation learning and security analytics, in: IEEE S&P, 2024, pp. 3590–3608.

[24] L. Yang et al., CADE: Detecting and explaining concept drift samples for security applications, in: USENIX Security Symposium, 2021, pp. 2327–2344.

[25] R. Vaarandi, A. Guerra-Manzanares, Network IDS alert classification with active learning techniques, Journal of Information Security and Applications 81 (2024) 103687.

[26] L. Chavali, A. Krishnan, P. Saxena, B. Mitra, A. Sreevallabh Chivukula, Off-policy actor-critic deep reinforcement learning for alert prioritization in intrusion detection systems, Computers & Security 142 (2024) 103854.

[27] I. Sharafaldin, A. H. Lashkari, A. A. Ghorbani, Toward generating a new intrusion detection dataset and intrusion traffic characterization, in: ICISSP, 2018, pp. 108–116.

[28] K. Hynek, J. Luxemburk, J. Pesek, T. Cejka, P. Siska, CESNET-TLS-Year22: A year-spanning TLS network traffic dataset from backbone lines, Scientific Data 11 (2024) 1156.

[29] N. Moustafa, J. Slay, UNSW-NB15: A comprehensive data set for network intrusion detection systems, in: Military Communications and Information Systems Conference (MilCIS), 2015, pp. 1–6.

[30] J. L. Guerra, C. Catania, E. Veas, Datasets are not enough: Challenges in labeling network traffic, Computers & Security 120 (2022) 102810.

[31] Y. Qing et al., Low-quality training data only? A robust framework for detecting encrypted malicious network traffic, arXiv preprint arXiv:2309.04798, 2023.

[32] Z. Zhao et al., Enhancing robustness of on-line learning models on highly noisy data, IEEE TDSC 18 (5) (2021) 2177–2192.

[33] N. Wang et al., FeCo: Boosting intrusion detection capability in IoT networks via contrastive learning, in: IEEE INFOCOM, 2022, pp. 1409–1418.

[34] Y. Yue et al., Contrastive learning enhanced intrusion detection, IEEE TNSM 19 (4) (2022) 4232–4247.

[35] Y. Liu et al., DeepDRAC: Disposition recommendation for alert clusters based on security event patterns, IEEE TIFS 20 (2025) 6443–6458.

[36] H. G. Shon, Y. Lee, M. Yoon, Semi-supervised alert filtering for network security, Electronics 12 (23) (2023) 4755.

[37] W. Wang et al., Transformer-based framework for alert aggregation and attack prediction in a multi-stage attack, Computers & Security 136 (2024) 103533.

[38] X. Xia et al., Sample selection with uncertainty of losses for learning with noisy labels, arXiv preprint arXiv:2106.00445, 2021.

[39] C. Liu et al., Noise-resistant deep metric learning with ranking-based instance selection, in: CVPR, 2021, pp. 6811–6820.

[40] S. Yang et al., RecDA: Concept drift adaptation with representation enhancement for network intrusion detection, in: ACM SIGKDD, 2024, pp. 3818–3828.

[41] G. Andresini et al., Insomnia: Towards concept-drift robustness in network intrusion detection, in: ACM AISec, 2021, pp. 111–122.

[42] K. Hynek et al., CESNET-TLS-Year22 dataset, Zenodo record 10608607, 2024.

[43] N. Moustafa, J. Slay, The evaluation of network anomaly detection systems: UNSW-NB15, Information Security Journal 25 (1–3) (2016) 18–31.